

Cambodia Daily Consumer Price Index (CPI)

End-to-End Data Lineage, Transformation Logic, and Column Dictionary

Medallion Architecture Specification (Bronze → Silver → Gold → Econometric Nowcast)

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Abstract

This technical specification provides the complete, auditable end-to-end data lineage, transformation mathematics, and column-level dictionary for the **Cambodia Daily Consumer Price Index (CPI) Medallion Data Platform**. The platform daily ingests multi-source unstructured and semi-structured retail data across 25 retail web scrapers and official government APIs, processing over 120,000 raw daily product observations into conformed UN COICOP 2018 analytical marts and econometric nowcasts. This document rigorously traces every table and column across the Medallion progression: from raw JSON extraction in **Bronze (Staging)**, through text normalization, vector cosine deduplication, and AI classification in **Silver**, up to Jevons micro-indices, Laspeyres macro synthesis, and 5-basket RidgeCV nowcasting in **Gold**. Each column is formally mapped with its source provenance, SQL transformation expression, mathematical formula, nullability constraint, and business definition.

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1 Medallion Architectural Lineage Overview

The Cambodia CPI data platform is architected as a strict Medallion data engineering pipeline operating on a single-writer, dependency-enforced DAG topology managed by Apache Airflow 2.9.3 and dbt Core.

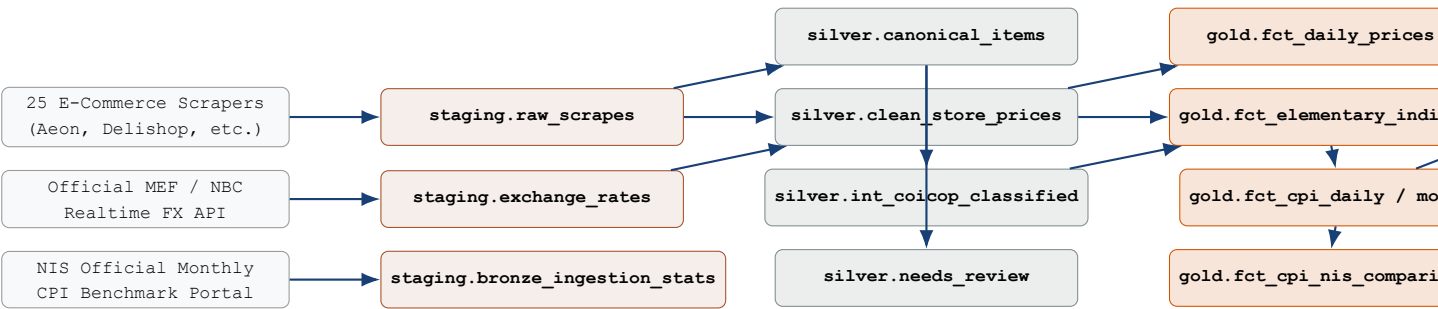


Figure 1: Complete End-to-End DAG Data Flow: From Raw Retail Sources to Analytical Fact Marts and Econometric Nowcasts.

1.1 Pipeline Execution Cadence & Single-Writer Architecture

Data flows sequentially across four Airflow DAGs executing on an ICT daily schedule:

1. **Scraper Ingestion Fan-Out (02:00–05:30 ICT):** 25 asynchronous Python scrapers extract raw listings and land JSON batches into `staging.raw_scrapes`.
2. **Ingestion Circuit Breaker (05:45 ICT):** `pipeline/circuit_breaker.py` evaluates price velocity deltas against 7-day rolling medians. Anomalous batches are quarantined in `ops.circuit_breaker_events`.
3. **Silver Curation DAG (06:00 ICT):** Normalizes strings, assigns 768-dimensional embeddings via Gemini, runs pgvector cosine clustering, executes 5-tier COICOP classification, and unifies dual-currency prices into KHR.
4. **Gold Synthesis & Nowcasting DAG (08:00 ICT):** Computes axiomatic Jevons micro-indices, aggregates Laspeyres 12-division indices, estimates expanding RidgeCV parameters, and executes the intra-month discrete arithmetic mean nowcast.

2 Bronze Staging Layer: Ingestion & Circuit Breakers

2.1 Table: `staging.raw_scrapes`

Description: Append-only raw landing table storing immutable batch payloads from scrapers.

Grain: One row per scrape execution batch per store.

Authoritative Writer: `pipeline/bronze_ingestion.py`.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
<code>scrape_id</code>	UUID	✓	No	<code>uuid.uuid4()</code>	Deterministic primary surrogate key for the scrape execution batch.
<code>store_slug</code>	<code>VARCHAR(50)</code>		No	Scraper metadata identifier	Unique identifier of the retail source (e.g. 'aeon', 'delishop').
<code>scrape_date</code>	DATE		No	Current date in Asia/Phnom_Penh	Accounting date for price validity.
<code>payload</code>	JSONB		No	HTTP / Browser JSON Array	Raw unparsed array of product dictionaries containing title, price, etc.
<code>record_count</code>	INTEGER		No	<code>len(payload)</code>	Total items harvested in this batch.
<code>created_at</code>	<code>TIMESTAMPZ</code>		No	<code>NOW() AT TIME ZONE 'UTC'</code>	Immutable UTC ingestion timestamp.

Table 1: Detailed Column Lineage for `staging.raw_scrapes`.

2.2 Table: `staging.exchange_rates`

Description: Daily official USD/KHR exchange rates for dual-currency normalization.

Grain: One row per calendar date.

Authoritative Writer: `pipeline/mef_fx_importer.py`.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
<code>execution_date</code>	DATE	✓	No	MEF API response date	Official publication date of the exchange rate.
<code>rate</code>	<code>NUMERIC(10,4)</code>		No	MEF rate (fallback: NBC/4044)	Khmer Riel (KHR) per 1.00 USD. Clamped to [3800, 4500].
<code>source_currency</code>	<code>VARCHAR(3)</code>		No	Constant 'USD'	Base quotation currency.
<code>target_currency</code>	<code>VARCHAR(3)</code>		No	Constant 'KHR'	Local transaction settlement currency.
<code>source_api</code>	<code>VARCHAR(50)</code>		No	Provider identifier	Provenance tag ('mef_api', 'nbc_api', 'fallback_static').
<code>fetches_at</code>	<code>TIMESTAMPZ</code>		No	<code>NOW()</code>	Audit timestamp of API extraction.

Table 2: Detailed Column Lineage for `staging.exchange_rates`.

2.3 Table: `ops.circuit_breaker_events`

Description: Automated security barrier logs preventing corrupt ingestion batches from entering Silver.

Grain: One row per evaluated store ingestion batch.

Authoritative Writer: pipeline/circuit_breaker.py.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
event_id	BIGSERIAL	✓	No	Database sequence	Unique auto-incrementing audit log identifier.
store_slug	VARCHAR(50)		No	raw_scrapes.store_slug	Retailer under evaluation.
scrape_date	DATE		No	raw_scrapes.scrape_date	Date under evaluation.
status	VARCHAR(20)		No	Evaluated decision	Gate outcome: 'PASSED', 'WARNING_FLAGGED', or 'HALTED'.
record_count	INTEGER		No	Batch record count	Number of observations in batch.
median_price_khr	NUMERIC(12,2)		No	Batch median price	Current day median item price normalized to KHR.
price_velocity_delta_pct	NUMERIC(6,2)		Yes	$\frac{Med_t - Med_{7d}}{Med_{7d}} \times 100\%$	Percentage drift from rolling 7-day median. Flags warning if > 15%.
created_at	TIMESTAMPTZ		No	NOW()	Evaluation timestamp.

Table 3: Detailed Column Lineage for ops.circuit_breaker_events.

3 Silver Layer: Normalization, Embeddings & Classification

3.1 Table: `silver.canonical_items`

Description: Central product catalog deduplicating distinct store listings into unified representative varieties.

Grain: One row per unique canonical consumer product.

Authoritative Writer: `pipeline/canonical_matcher.py`.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
<code>item_id</code>	UUID	✓	No	<code>uuid5(DNS, clean_name)</code>	Master product surrogate key.
<code>canonical_name</code>	TEXT		No	Cleaned string from <code>pipeline/text_clean.py</code>	Deterministically normalized product name: uppercase, stripped of promo noise and diacritics.
<code>brand</code>	VARCHAR(100)		Yes	Regex extraction / payload	Identified brand entity (e.g. 'COCA COLA', 'NESTLE').
<code>package_size</code>	TEXT		Yes	Regex extraction	Raw normalized size descriptor (e.g. '500g', '1L').
<code>unit_value</code>	NUMERIC(10,3)		Yes	Numerical regex parse	Standardized quantity magnitude (e.g. 500.0).
<code>unit_metric</code>	VARCHAR(10)		Yes	Unit extraction regex	Standardized measurement metric ('g', 'kg', 'ml', 'l').
<code>match_method</code>	VARCHAR(30)		No	Resolution engine tag	Discovery lineage: 'exact_match', 'embedding_similarity', or 'new_item'.
<code>match_confidence</code>	NUMERIC(5,4)		No	Cosine score: $1 - \text{dist}$	Vector match similarity score $\in [0.0000, 1.0000]$. Threshold ≥ 0.8500 .
<code>created_at</code>	TIMESTAMPTZ		No	<code>NOW()</code>	Item registration timestamp.

Table 4: Detailed Column Lineage for `silver.canonical_items`.

3.2 Hierarchical Precedence: `silver.int_coicop_classified`

Description: Intermediate dbt transformation resolving UN COICOP 2018 classification codes.

Transformation Macro: `dbt/macros/coicop_classify_macro.sql`.

5-Tier Classification Resolution Precedence

The COICOP code for an item is resolved via a strict SQL CASE precedence cascade:

1. **Tier 1: Exact Overrides (coicop_override.csv):** Deterministic overrides for verified ambiguous products.
2. **Tier 2: Store Purity Mapping:** Mono-category stores mapped directly (e.g. PPWSA → Division 04 Water, EDC → Division 04 Electricity, Khmer24 Real Estate → Division 04 Rent).
3. **Tier 3: Critical Regex Traps (coicop_critical_traps.csv):** Word-boundary regex using PostgreSQL \y operators to prevent false positives (e.g. \ytoothpaste\y → 12, \ymotorcycle\y → 07).
4. **Tier 4: Gemini AI Drill-Down Cache (silver.dim_coicop_ai_cache):** High-confidence LLM classification using structured JSON outputs memoized permanently in PostgreSQL.
5. **Tier 5: Category Map Fallback (category_mapping.csv):** Baseline mapping from retailer breadcrumb taxonomy.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
item_id	UUID	✓	No	silver.canonical_items.item_id	Foreign key referencing master canonical item.
coicop_division	CHAR(2)		No	Resolved division code	2-digit UN COICOP Division ('01' to '12').
coicop_code	VARCHAR(10)		No	Resolved full classification	4-digit or 5-digit COICOP category code (e.g. '01.1.1.1').
coicop_method	VARCHAR(30)		No	Resolution engine step	Audit tag: 'purity_store', 'trap_rule', 'gemini_ai', etc.
confidence_score	NUMERIC(5,4)		No	Model / rule score	Categorization confidence. Trap rules = 1.0000, AI = [0.80, 0.99].

Table 5: Detailed Column Lineage for silver.int_coicop_classified.

3.3 Table: `silver.clean_store_prices`

Description: Cleaned, typed, and normalized daily price observations ready for economic index synthesis.

Grain: One row per scraped product observation.

Authoritative Writer: `dbt/models/silver/clean_store_prices.sql`.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
raw_price_id	UUID	✓	No	staging.raw_scrapes.pay-load	Unique identifier for the individual scraped listing.
scrape_date	DATE		No	staging.raw_scrapes.scrape_date	Observation date in ICT.
store_slug	VARCHAR(50)		No	staging.raw_scrapes.store_id	Retailer identifier.
item_id	UUID		No	silver.canonical_items.item_id	Foreign key to matched canonical product.
coicop_division	CHAR(2)		No	silver.int_coicop_classification	Assigned UN COICOP Division code.
price_raw	NUMERIC(14,2)		No	Extracted listing price	Raw parsed price in retailer's quotation currency.
currency	VARCHAR(3)		No	Extracted currency code	Store transaction currency: 'USD' or 'KHR'.
exchange_rate	NUMERIC(10,4)		No	staging.exchange_rates.rate	Official MEF USD/KHR exchange rate applied for normalization.
price_khr	NUMERIC(14,2)		No	IF currency = 'USD' THEN price_raw * rate ELSE price_raw	Primary Conformed Metric: Price normalized into Khmer Riel (KHR) to prevent downstream currency bias.
unit_price_khr	NUMERIC(14,2)		Yes	price_khr / unit_value	Standardized unit price (e.g. KHR per 1kg, 1L, or 1 piece).
original_price_khr	NUMERIC(14,2)		Yes	original_price_raw * rate (if USD)	Pre-discount list price in KHR.
discount_pct	NUMERIC(5,2)		No	clamp((orig - price) / orig * 100, 0, 95)	Promotional discount rate percentage.
on_promo	BOOLEAN		No	discount_pct > 0.0	Boolean flag indicating promotional discounting.
is_outlier	BOOLEAN		No	Hadi / Tukey IQR check	Set to TRUE if relative price movement exceeds 3.0 IQR within division.
is_fallback	BOOLEAN		No	Scraper status	Set to TRUE if observation originated from static fallback catalog.
fallback_reason	TEXT		Yes	Scraper exception log	Cause of fallback engagement (e.g. 'px_captcha_403').

Table 6: Detailed Column Lineage for silver.clean_store_prices.

4 Gold Layer: Axiomatic Micro-Indices & Macro Aggregation

4.1 Table: gold.fct_daily_prices

Description: Star-schema daily fact table aggregated at store-item grain for Business Intelligence.

Grain: (scrape_date, store_slug, item_id).

Authoritative Writer: dbt/models/gold/fct_daily_prices.sql.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
scrape_date	DATE	✓	No	clean_store_prices.scrape_date	Grain dimension: calendar scrape date.
store_slug	VARCHAR(50)	✓	No	clean_store_prices.store_slug	Grain dimension: retail store identifier.
item_id	UUID	✓	No	clean_store_prices.item_id	Grain dimension: canonical product key.
price_khr	NUMERIC(14,2)		No	$\text{round}\left(\exp\left(\frac{1}{N} \sum \ln P_{\text{khr}}\right), 2\right)$	Geometric mean of shelf prices for this item within this store.
unit_price_khr	NUMERIC(14,2)		Yes	$\text{round}\left(\exp\left(\frac{1}{N} \sum \ln P_{\text{unit}}\right), 2\right)$	Geometric mean of unit prices.
original_price_khr	NUMERIC(14,2)		Yes	MIN(original_price_khr)	Minimum baseline original price.
discount_pct	NUMERIC(5,2)		No	MAX(discount_pct)	Maximum promotional discount recorded.
on_promo	BOOLEAN		No	bool_or(on_promo)	Boolean indicator: product discounted on this date.
observation_count	INTEGER		No	COUNT(*)	Raw observations collapsed into this grain.
is_outlier	BOOLEAN		No	bool_or(is_outlier)	True if any observation was flagged as an outlier.
is_fallback	BOOLEAN		No	bool_or(is_fallback)	True if observation engaged a baseline fallback.

Table 7: Detailed Column Lineage for gold.fct_daily_prices.

4.2 Table: gold.fct_elementary_indices

Description: Jevons geometric micro-indices computed across canonical items relative to base month.

Grain: (calculation_date, item_id).

Authoritative Writer: pipeline/cpi_calculator.py (CPICalculationEngine.calculate_daily_elementary_ind

The Jevons Axiomatic Micro-Index Formula

In accordance with UN/ILO international guidelines, elementary item aggregates are computed using the unweighted geometric mean of cross-store price relatives:

$$R_{i,t} = \frac{P_{i,t}}{P_{i,0}} = \frac{\exp\left(\frac{1}{S_i} \sum_{s=1}^{S_i} \ln P_{i,s,t}\right)}{P_{i,0}} \quad (1)$$

where $P_{i,s,t}$ is the clean KHR price of item i in store s on day t , and $P_{i,0}$ is the fixed baseline anchor price established in August 2026.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
calculation_date	DATE	✓	No	Target computation date	Calendar date of the index calculation.
item_id	UUID	✓	No	clean_store_prices.item_id	Canonical item evaluated.
coicop_division	CHAR(2)		No	clean_store_prices.coicop_division	UN COICOP Division code.
base_price_khr	NUMERIC(14,2)		No	Base period August 2026 mean	Fixed baseline price $P_{i,0}$ in Khmer Riel.
current_price_khr	NUMERIC(14,2)		No	Cross-store geometric mean	Today's realized or imputed shelf price $P_{i,t}$.
price_ratio	NUMERIC(10,6)		No	$P_{i,t}/P_{i,0}$	Axiomatic Jevons Elementary Relative $R_{i,t}$.
is_imputed	BOOLEAN		No	Imputation status	True if item was unobserved today and price was imputed via division drift.
created_at	TIMESTAMPTZ		No	NOW()	Processing timestamp.

Table 8: Detailed Column Lineage for gold.fct_elementary_indices.

4.3 Table: gold.fct_cpi_daily & gold.fct_cpi_monthly

Description: Conformed 12-Division, Core, and National Headline Consumer Price Indices.

Grain: (calculation_date / cpi_month, coicop_division).

Authoritative Writer: pipeline/cpi_calculator.py (CPICalculationEngine.save_monthly_cpi).

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
cpi_month	DATE	✓	No	date_trunc('month', date)	First calendar day of the index month (YYYY-MM-01).
coicop_division	CHAR(2)	✓	No	Division code	2-digit UN COICOP Division ('01' to '12').
division_name	VARCHAR(100)		No	gold.coicop_weights.name	Official UN division classification title.
weight	NUMERIC(6,4)		No	Official NIS 2018 basket weight	Division expenditure weight w_j ($\sum w_j = 1.0000$).
monthly_division_index	NUMERIC(10,4)		No	$\exp\left(\frac{1}{K_j} \sum \ln R_{i,j}\right) \times 100$	Aggregated monthly index level for division j (2026 = 100).
monthly_headline_cpi	NUMERIC(10,4)		No	$\sum_{j=1}^{12} w_j \cdot I_{j,\text{month}}$	National Headline CPI aggregated across all 12 divisions.
monthly_core_cpi	NUMERIC(10,4)		No	$\frac{\sum_{j \notin \{01,04,07\}} w_j \cdot I_j}{\sum w_j}$	National Core CPI excluding Food, Housing Utilities, and Fuels.
mom_inflation_pct	NUMERIC(8,4)		Yes	$\frac{I_{j,m} - I_{j,m-1}}{I_{j,m-1}} \times 100\%$	Month-over-Month percentage change for division j .
yoy_inflation_pct	NUMERIC(8,4)		Yes	$\frac{I_{j,m} - I_{j,m-12}}{I_{j,m-12}} \times 100\%$	Year-over-Year percentage change for division j .
headline_mom_inflation_pct	NUMERIC(8,4)		Yes	$\frac{I_{\text{head},m} - I_{\text{head},m-1}}{I_{\text{head},m-1}} \times 100\%$	National Headline Month-over-Month inflation rate.
headline_yoy_inflation_pct	NUMERIC(8,4)		Yes	$\frac{I_{\text{head},m} - I_{\text{head},m-12}}{I_{\text{head},m-12}} \times 100\%$	National Headline Year-over-Year inflation rate.
item_count	INTEGER		No	COUNT(DISTINCT item_id)	Distinct rep-

5 Econometric Nowcasting Marts: Exact Lineage & Formulas

5.1 Table: `gold.nowcast_calibrated_parameters`

Description: Stores macroeconomic empirical elasticities (β_k) and optimal L_2 shrinkage penalties (λ^*) estimated via 36-month expanding RidgeCV.

Grain: One row per calibration run.

Authoritative Writer: `ml/calibration.py` (`RidgeCalibrationEngine.persist_parameters`).

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
<code>calibration_date</code>	DATE	✓	No	Run date	Execution date of the RidgeCV expanding panel optimization.
<code>sample_start</code>	DATE		No	First panel month	Earliest historical month in the estimation window.
<code>sample_end</code>	DATE		No	Last panel month	Most recent completed month in the estimation panel.
<code>sample_months</code>	INTEGER		No	Panel length	Total monthly observations evaluated (M).
<code>alpha_drift</code>	NUMERIC(8, 6)		No	RidgeCV intercept $\hat{\alpha}$	Empirical monthly base drift intercept ($\hat{\alpha} = +0.001899$).
<code>lambda_penalty</code>	NUMERIC(8, 4)		No	Optimal λ^* via LOOCV	Regularization penalty chosen from $[0.01, 1000.0]$ ($\lambda^* = 0.0100$).
<code>beta_food</code>	NUMERIC(6, 4)		No	Ridge coefficient β_1	Pass-through elasticity for Division 01 (Food, 44.8% weight).
<code>beta_alcohol</code>	NUMERIC(6, 4)		No	Ridge coefficient β_2	Pass-through elasticity for Division 02 (Alcohol, 1.6% weight).
<code>beta_housing</code>	NUMERIC(6, 4)		No	Ridge coefficient β_4	Pass-through elasticity for Division 04 (Housing Utilities, 17.1% weight).
<code>beta_transport</code>	NUMERIC(6, 4)		No	Ridge coefficient β_7	Pass-through elasticity for Division 07 (Transport, 12.2% weight).
<code>beta_restaurant</code>	NUMERIC(6, 4)		No	Ridge coefficient β_{11}	Pass-through elasticity for Division 11 (Restaurants, 5.9% weight).
<code>beta_fx</code>	NUMERIC(6, 4)		No	Ridge coefficient β_{fx}	Empirical pass-through elasticity for USD/KHR exchange rate movements.
<code>oos_relative_rmse</code>	NUMERIC(6, 4)		Yes	$RMSE_{\text{model}}/RMSE_{\text{RW}}$	Out-of-sample performance vs. Atkeson-Ohanian Random Walk (0.8232).
<code>oos_r2</code>	NUMERIC(6, 4)		Yes	Out-of-sample R^2	Predictive variance explained on out-of-sample evaluation split.

Table 10: Detailed Column Lineage for `gold.nowcast_calibrated_parameters`.

5.2 Table: `gold.fct_cpi_nowcast`

Description: Real-time daily Month-to-Date inflation nowcasts produced at 08:00 AM ICT.

Grain: (`nowcast_date`, `target_month`, `model_name`).

Authoritative Writer: `ml/nowcaster.py` (`CPINowcaster.save_nowcast`).

The Exact Discrete Arithmetic Mean Projection

In the production pipeline (`ml/nowcaster.py`), the projected remaining price index for unobserved days $k \in [1, N]$ is calculated using the **Exact Discrete Arithmetic Mean**:

$$\bar{I}_{j,\text{rem}}(d) = \frac{1}{N} \sum_{k=1}^N I_{j,d} \cdot \exp(\mu_j(d) \cdot k) \quad (2)$$

implemented as `np.mean(latest_div * np.exp(div_drift * future_days))`. Microdata prices are already aggregated into geometric Jevons indices in Gold, filtering out idiosyncratic store churn.

Target Column	Data Type	PK	Null?	Source Column / Expr	Business Definition & Logic
nowcast_date	DATE	✓	No	Run date	Date the nowcast was generated (daily at 08:00 ICT). Month being nowcasted (YYYY-MM-01). Econometric model version: 'hybrid_ridge_5basket_ Number of realized elapsed calendar days with observed scraped prices. Unobserved future days remaining until month-end. Total calendar days in target month (28, 29, 30, or 31). Arithmetic average of realized daily headline CPI up to day t. Projected average index for remaining unobserved days $d \in [t + 1, T]$. Core Business Indicator: Projected full Month-over-Month inflation rate. Projected final Month-End Headline CPI (2026 =
target_month	DATE	✓	No	First day of forecast month	
model_name	VARCHAR(50)	✓	No	Constant tag	
days_observed	INTEGER		No	Day of month (t)	
days_remaining	INTEGER		No	$N = T - t$	
days_in_month	INTEGER		No	Month length (T)	
realized_cpi_so_far	NUMERIC(10, 4)		No	$\frac{1}{t} \sum_{d=1}^t I_{\text{head}, d}$	
projected_remaining_cpi	NUMERIC(10, 4)		No	Exact discrete mean	
projected_mom_pct	NUMERIC(8, 4)		No	$\frac{\hat{I}_T - I_{m-1}}{I_{m-1}} \times 100\%$	
nowcast_headline_cpi	NUMERIC(10, 4)		No	Final synthesized index	

6 Master Column-Level Lineage Matrix

The following master matrix cross-references every active warehouse column with its direct upstream source table, column dependency, transformation formula, and functional role:

Target Table	Target Column	Upstream Source	Source Column	Transformation / Rule
staging.raw_scrapes	payload	Retail Scraper	HTTP Response	Raw JSON string parse
staging.exchange_rates	rate	MEF Realtime API	API JSON	Value parse with NB
ops.circuit_breaker_events	price_velocity_delta_pct	clean_store_prices	price_khr	$(Med_t - Med_{7d})/Med_t$
silver.canonical_items	canonical_name	staging.raw_scrapes	payload->>'title'	pipeline/text_clean
silver.canonical_items	match_confidence	gemini-embedding-2	Text Vector	$1 - CosineDistance$
silver.int_coicop_classified	coicop_division	Multiple Seeds / AI	Keyword / Vector	5-Tier SQL Precedence
silver.clean_store_prices	price_khr	staging.raw_scrapes	price_raw, currency	$P_{raw} \cdot Rate_{USD/KHR}$ if
silver.clean_store_prices	unit_price_khr	clean_store_prices	price_khr, unit_value	$P_{khr}/unit_value$
silver.clean_store_prices	discount_pct	clean_store_prices	orig_khr, price_khr	$clamp((P_0 - P)/P_0)$
silver.clean_store_prices	is_outlier	clean_store_prices	price_khr	Hadi / Tukey IQR (>
gold.fct_daily_prices	price_khr	clean_store_prices	price_khr	$\exp(\frac{1}{N} \sum \ln P_{khr})$
gold.fct_elementary_indices	price_ratio	clean_store_prices	price_khr	$P_{i,t}/P_{i,0}$
gold.fct_elementary_indices	is_imputed	clean_store_prices	Scrape status	Missing check → D

7 Operational Governance & Data Quality Assurance

7.1 dbt Test Suites and Invariant Assertions

To preserve data integrity across the lineage chain, the repository executes automated dbt tests on every pipeline cycle:

- **Uniqueness & Non-Null Invariants:** All surrogate primary keys (`item_id`, `scrape_id`, `raw_price_id`) and partition dates (`scrape_date`, `calculation_date`, `cpi_month`) strictly assert `not_null` and `unique`.
- **Axiomatic Weight Closure:** The expenditure weights in `gold.cambodia_cpi_coicop_weights_breakdown` and `gold.coicop_weights` assert $\sum w_j = 1.0000 \pm 0.0001$.
- **Positive Price Invariant:** All prices assert $P_{\text{KHR}} > 0$. Zero or negative values are automatically quarantined in `silver.needs_review`.
- **Classification Completeness Invariant:** The count of items with `coicop_division IS NULL` or `coicop_code LIKE '%.unclassified'` in `silver.canonical_items` must strictly equal 0.

7.2 Data Freshness SLAs

- **Bronze Freshness:** Warn after 24 hours, error after 48 hours for `staging.raw_scrapes` and `staging.exchange_rates`.
- **Gold Macro SLA:** `gold.fct_cpi_daily` and `gold.fct_cpi_nowcast` must complete publication by 08:30 AM ICT daily to provide timely economic signals for open-market operations at the National Bank of Cambodia.

Technical Specification Summary

This document forms the official reference baseline for all data modeling, database audits, and econometric nowcasting operations within the Cambodia CPI Pipeline. Any schema alterations or upstream column modifications must update this specification and pass the automated dbt and pytest verification suites.